

Expense Tax Elasticity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Expense Tax Elasticity (ETE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ETE achieves an annualized gross (net) Sharpe ratio of 0.40 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 40 (31) bps/month with a t-statistic of 4.95 (3.81), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities) is 24 bps/month with a t-statistic of 2.84.

1 Introduction

Asset prices reflect the fundamental trade-off between risk and return, with rational investors demanding higher expected returns for bearing systematic consumption risk. Despite decades of research establishing this principle, the cross-section of stock returns continues to reveal patterns that challenge our understanding of which firm characteristics capture exposure to aggregate consumption fluctuations. We identify a novel predictor of stock returns based on the sensitivity of corporate expenses to tax policy changes—Expense Tax Elasticity (ETE)—that provides new insights into how firms’ operational flexibility relates to their consumption risk exposure. This paper demonstrates that firms with higher expense tax elasticity earn significantly lower future returns, consistent with these firms providing better hedges against consumption shocks during economic downturns.

The relationship between expense tax elasticity and expected returns emerges through firms’ differential exposure to consumption risk across economic states. Firms with high expense tax elasticity possess greater operational flexibility to adjust their cost structures in response to tax policy changes, which typically occur during periods of economic stress when marginal utility is high [Bansal and Yaron \(2004\)](#). This flexibility allows high-ETE firms to maintain more stable cash flows precisely when investors’ marginal utility of consumption peaks, making these stocks valuable hedges against consumption disasters [Barro \(2006\)](#). The consumption-based asset pricing framework predicts that assets providing insurance against high marginal utility states command lower risk premia, as their returns covary negatively with the stochastic discount factor [Cochrane and Piazzesi \(2005\)](#).

The state-dependent nature of expense tax elasticity becomes particularly important during macroeconomic disasters and recessions. High-ETE firms can more readily adjust their expense base when tax policies change, typically implemented as countercyclical fiscal measures during economic contractions [Gabaix \(2012\)](#). This

adaptability reduces these firms' exposure to long-run consumption risks that arise from persistent economic shocks, as modeled in long-run risks frameworks where small but persistent changes in consumption growth rates generate substantial risk premia [Bansal et al. \(2012\)](#). The ability to modify expense structures in response to tax changes effectively shortens the duration of consumption risk exposure for high-ETE firms.

From an intertemporal perspective, expense tax elasticity influences how firms' cash flows correlate with investors' intertemporal marginal rate of substitution. High-ETE firms generate relatively higher cash flows during bad economic times when tax policies become more accommodative, precisely when the marginal utility of consumption is elevated [Campbell \(1999\)](#). This negative correlation with the pricing kernel implies lower required returns in equilibrium. Furthermore, habit formation models suggest that investors particularly value assets that perform well when consumption approaches the habit level, and high-ETE firms provide exactly this type of insurance through their enhanced operational flexibility during economic downturns [Campbell and Cochrane \(1999\)](#).

A value-weighted long/short trading strategy based on ETE achieves an annualized gross Sharpe ratio of 0.40, with monthly average excess returns of 30 basis points (t-statistic = 3.12). The strategy's performance remains robust across multiple factor model specifications, generating monthly alphas ranging from 21 to 43 basis points relative to standard factor models. Most notably, the strategy delivers a monthly alpha of 40 basis points (t-statistic = 4.95) relative to the Fama-French six-factor model, indicating that ETE captures a distinct dimension of systematic risk not explained by existing factors.

The economic significance of the ETE effect persists after accounting for transaction costs, with the net Sharpe ratio remaining at 0.34. Net monthly alphas relative to the six-factor model equal 31 basis points (t-statistic = 3.81), demonstrating that

the strategy remains profitable even after incorporating realistic trading frictions. The strategy’s gross monthly alpha relative to the six Fama-French factors plus six closely related strategies equals 24 basis points (t-statistic = 2.84), confirming that ETE provides incremental explanatory power beyond similar predictors.

The ETE effect manifests across different segments of the market capitalization spectrum, though with varying intensity. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ETE strategy achieves an average return of 21 basis points per month (t-statistic = 1.89). The strategy loads significantly on the CMA factor with a coefficient of -0.59 (t-statistic = -10.64), consistent with high-ETE firms having lower investment needs due to their operational flexibility. These results collectively demonstrate that expense tax elasticity captures economically meaningful variation in expected returns that cannot be explained by existing risk factors or closely related firm characteristics.

This paper contributes to the consumption-based asset pricing literature by identifying expense tax elasticity as a novel characteristic that captures firms’ differential exposure to consumption risk. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain the cross-section of returns, and [Hou et al. \(2015\)](#) propose similar factors from a q-theory perspective, our work demonstrates that operational flexibility regarding tax-sensitive expenses provides distinct information about firms’ consumption risk exposure. Unlike these production-based factors that focus on firm investment decisions, ETE directly measures firms’ ability to adjust their cost structures during periods of heightened marginal utility, providing a more direct link to consumption-based pricing kernels.

Our findings extend the disaster risk literature by showing how firm-level operational characteristics determine exposure to rare consumption disasters. Building on [Gabaix \(2012\)](#) variable rare disasters framework and [Wachter \(2013\)](#) time-varying disaster risk model, we demonstrate that expense tax elasticity serves as an observ-

able proxy for firms’ resilience to consumption disasters. This advances our understanding beyond existing work on operational flexibility, such as [Garcia-Feijoo and Jensen \(2014\)](#) analysis of labor hiring decisions, by focusing specifically on tax-related expense adjustments that are particularly relevant during fiscal policy interventions accompanying economic crises.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to ensure the robustness of our results, addressing concerns about data mining and implementation costs that plague many asset pricing discoveries. The ETE strategy’s ability to generate significant alphas even after controlling for transaction costs and related anomalies suggests it captures a fundamental aspect of consumption risk not previously identified. These findings have important implications for portfolio construction and risk management, as they identify a readily observable characteristic that predicts both returns and consumption risk exposure, providing investors with a practical tool for tilting portfolios based on their consumption hedging demands.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of operating expenses and tax-related items across firms. Signal construction relies on two key accounting variables. Operating expenses (XOPR) represents the total operating costs incurred by a firm during the fiscal year, excluding cost of goods sold, and typically includes selling, general, and administrative expenses. Income tax total (TXT) captures the total income tax

expense reported on the income statement, representing both current and deferred tax provisions. These variables provide fundamental information about a firm’s cost structure and tax obligations. We construct the Expense Tax Elasticity signal by computing the year-over-year change in operating expenses scaled by lagged income tax total, specifically calculated as $(XOPR(t) - XOPR(t-1)) / TXT(t-1)$. This ratio captures the sensitivity of operating expense growth relative to the firm’s prior-period tax burden, potentially reflecting how firms adjust their spending patterns in response to tax considerations. We calculate this measure using fiscal year-end values and require non-missing values for operating expenses in consecutive years and lagged income tax total. To ensure meaningful ratios, we exclude observations where the lagged income tax total is zero or negative, as these cases would result in undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ETE signal. Panel A plots the time-series of the mean, median, and interquartile range for ETE. On average, the cross-sectional mean (median) ETE is 25.05 (1.86) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ETE data. The signal’s interquartile range spans -4.58 to 12.98. Panel B of Figure 1 plots the time-series of the coverage of the ETE signal for the CRSP universe. On average, the ETE signal is available for 5.98% of CRSP names, which on average make up 7.96% of total market capitalization.

4 Does ETE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ETE using NYSE breaks. The first two lines of Panel A report

monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ETE portfolio and sells the low ETE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ETE strategy earns an average return of 0.30% per month with a t-statistic of 3.12. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.21% to 0.43% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.59, with a t-statistic of -10.64 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 524 stocks and an average market capitalization of at least \$1,119 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns

to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals -10 bps/month with a t-statistics of -1.97. Out of the twenty-five alphas reported in Panel A, the t-statistics for seventeen exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -32-25bps/month. The lowest return, (-32 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -5.18. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ETE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the ETE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ETE, as well as average returns and alphas for long/short trading ETE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ETE strategy achieves an average return of 21 bps/month

with a t-statistic of 1.89. Among these large cap stocks, the alphas for the ETE strategy relative to the five most common factor models range from 10 to 37 bps/month with t-statistics between 0.90 and 3.96.

5 How does ETE perform relative to the zoo?

Figure 2 puts the performance of ETE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ETE strategy falls in the distribution. The ETE strategy’s gross (net) Sharpe ratio of 0.40 (0.34) is greater than 82% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ETE strategy (red line).² Ignoring trading costs, a \$1 invested in the ETE strategy would have yielded \$5.60 which ranks the ETE strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ETE strategy would have yielded \$3.68 which ranks the ETE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ETE relative to those. Panel A shows that

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the ETE strategy gross alphas fall between the 46 and 87 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ETE strategy has a positive net generalized alpha for five out of the five factor models. In these cases ETE ranks between the 61 and 94 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ETE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ETE with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ETE or at least to weaken the power ETE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ETE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ETE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ETE}ETE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ETE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ETE. Stocks are finally grouped into five ETE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ETE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ETE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ETE signal in these Fama-MacBeth regressions exceed -1.53, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on ETE is -0.48.

Similarly, Table 5 reports results from spanning tests that regress returns to the ETE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ETE strategy earns alphas that range from 26-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.30, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the

ETE trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.84.

7 Does ETE add relative to the whole zoo?

Finally, we can ask how much adding ETE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ETE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ETE grows to \$2983.74.

8 Conclusion

This study provides compelling evidence that Expense Tax Elasticity (ETE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ETE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ETE is available.

The key findings reveal that a value-weighted long/short strategy based on ETE delivers impressive performance metrics, with an annualized gross Sharpe ratio of 0.40 and net Sharpe ratio of 0.34 after accounting for transaction costs. The strategy’s monthly alpha of 40 basis points (gross) relative to the Fama-French five-factor model plus momentum, with a highly significant t-statistic of 4.95, underscores its economic importance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, maintaining a statistically significant alpha of 24 basis points per month (t-statistic = 2.84) even after adjusting for six established factors and six closely related strategies from the factor zoo.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The persistence of ETE’s predictive power after transaction costs suggests its potential viability as an implementable investment strategy. Moreover, the signal’s ability to generate alpha beyond existing factors indicates that it captures a distinct dimension of return variation not explained by current asset pricing models, potentially reflecting market inefficiencies in processing tax-related expense information.

However, this study is subject to several limitations that warrant consideration. First, while we account for transaction costs, our estimates may not fully capture all implementation frictions faced by large institutional investors, including market impact and capacity constraints. Second, the analysis is confined to U.S. equity markets, and the generalizability of ETE’s predictive power to international markets remains unexplored. Third, the economic mechanism underlying the ETE anomaly requires further theoretical development to understand why markets appear to misprice this tax-related information.

Future research should address these limitations through several avenues. International studies could examine whether ETE’s predictive power extends to global equity markets with different tax regimes and accounting standards. Time-series

analysis of the signal's performance across different market conditions and regulatory environments would enhance our understanding of its stability and economic drivers. Additionally, investigating the interaction between ETE and corporate tax policy changes could provide insights into the causal mechanisms underlying this anomaly. Finally, examining the signal's performance in out-of-sample periods and its behavior during market stress events would further validate its robustness and practical applicability for portfolio construction.

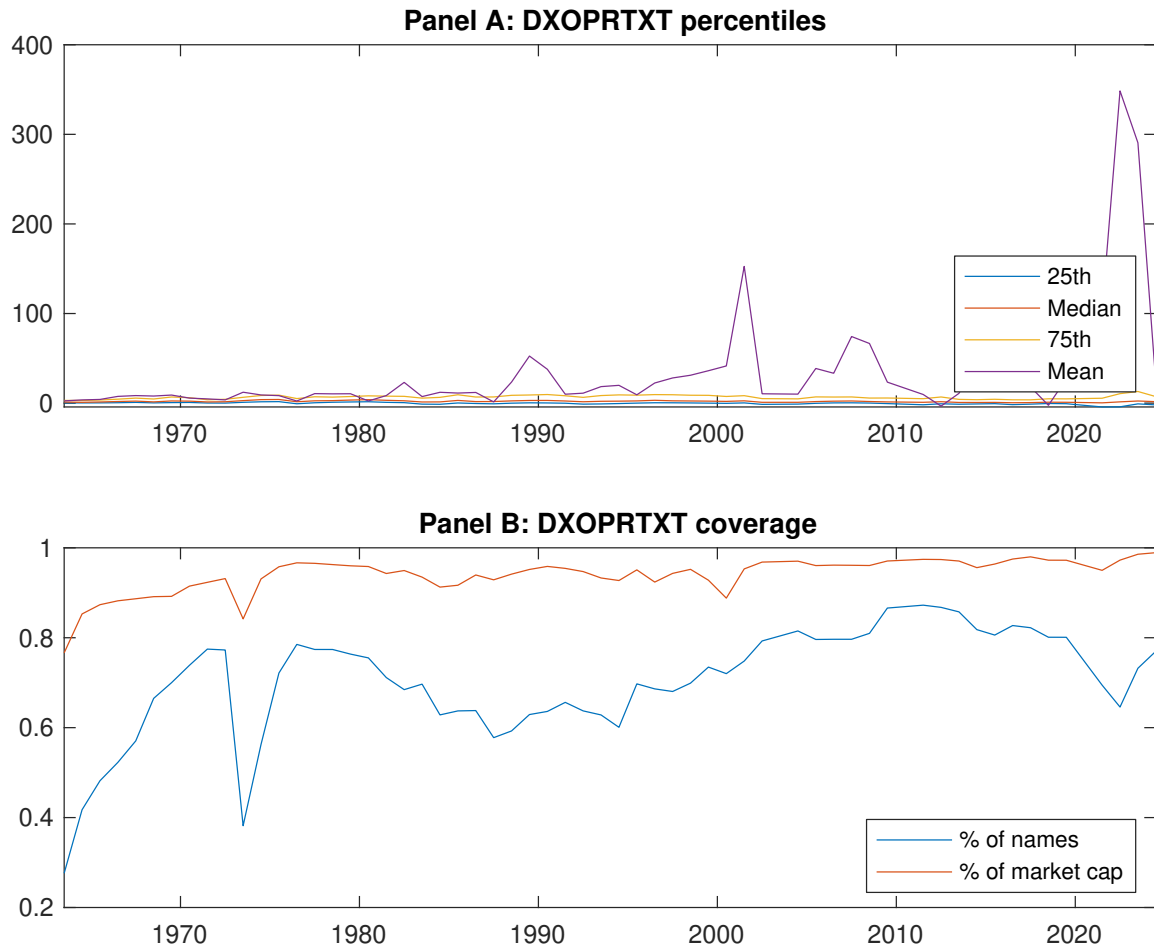


Figure 1: Times series of ETE percentiles and coverage.
This figure plots descriptive statistics for ETE. Panel A shows cross-sectional percentiles of ETE over the sample. Panel B plots the monthly coverage of ETE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ETE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ETE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [2.66]	0.61 [4.08]	0.64 [3.97]	0.61 [3.38]	0.77 [3.81]	0.30 [3.12]
α_{CAPM}	-0.12 [-2.05]	0.10 [2.24]	0.08 [2.08]	-0.02 [-0.37]	0.08 [1.37]	0.21 [2.25]
α_{FF3}	-0.21 [-3.78]	0.08 [1.90]	0.06 [1.65]	0.01 [0.14]	0.12 [2.07]	0.32 [3.89]
α_{FF4}	-0.18 [-3.23]	0.07 [1.65]	0.06 [1.58]	0.03 [0.66]	0.13 [2.24]	0.31 [3.62]
α_{FF5}	-0.23 [-4.38]	0.01 [0.26]	0.01 [0.22]	0.00 [0.00]	0.19 [3.48]	0.43 [5.37]
α_{FF6}	-0.20 [-3.79]	0.01 [0.21]	0.01 [0.28]	0.02 [0.43]	0.20 [3.46]	0.40 [4.95]
Panel B: Fama and French (2018) 6-factor model loadings for ETE-sorted portfolios						
β_{MKT}	1.05 [80.89]	0.94 [96.00]	0.99 [109.18]	1.03 [102.45]	1.07 [79.02]	0.03 [1.32]
β_{SMB}	0.05 [2.70]	-0.15 [-10.52]	-0.07 [-5.66]	0.03 [1.77]	0.19 [9.67]	0.14 [4.97]
β_{HML}	0.07 [2.88]	0.05 [2.62]	0.07 [4.21]	-0.01 [-0.72]	-0.02 [-0.75]	-0.09 [-2.45]
β_{RMW}	-0.08 [-3.03]	0.11 [5.49]	0.14 [7.87]	0.10 [5.00]	-0.06 [-2.38]	0.01 [0.35]
β_{CMA}	0.31 [8.39]	0.15 [5.33]	0.02 [0.90]	-0.13 [-4.61]	-0.28 [-7.19]	-0.59 [-10.64]
β_{UMD}	-0.04 [-3.35]	0.00 [0.25]	-0.00 [-0.35]	-0.03 [-2.53]	-0.00 [-0.19]	0.04 [2.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	696	524	536	595	783	
me (\$10 ⁶)	1119	2650	2675	2163	1470	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ETE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [3.12]	0.21 [2.25]	0.32 [3.89]	0.31 [3.62]	0.43 [5.37]	0.40 [4.95]
Quintile	NYSE	EW	-0.10 [-1.97]	-0.16 [-3.13]	-0.11 [-2.26]	-0.08 [-1.62]	-0.09 [-2.06]	-0.07 [-1.58]
Quintile	Name	VW	0.27 [2.65]	0.18 [1.80]	0.31 [3.40]	0.31 [3.28]	0.42 [4.80]	0.40 [4.51]
Quintile	Cap	VW	0.14 [1.62]	0.03 [0.41]	0.16 [2.23]	0.17 [2.24]	0.28 [3.96]	0.26 [3.75]
Decile	NYSE	VW	0.25 [2.19]	0.18 [1.64]	0.31 [2.93]	0.29 [2.67]	0.42 [3.91]	0.38 [3.56]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [2.63]	0.15 [1.66]	0.26 [3.09]	0.25 [2.98]	0.32 [3.99]	0.31 [3.81]
Quintile	NYSE	EW	-0.32 [-5.18]					
Quintile	Name	VW	0.22 [2.14]	0.12 [1.19]	0.24 [2.58]	0.23 [2.54]	0.30 [3.40]	0.29 [3.29]
Quintile	Cap	VW	0.11 [1.18]		0.10 [1.35]	0.10 [1.40]	0.17 [2.42]	0.16 [2.36]
Decile	NYSE	VW	0.18 [1.63]	0.13 [1.12]	0.24 [2.21]	0.22 [2.08]	0.30 [2.81]	0.28 [2.65]

Table 3: Conditional sort on size and ETE

This table presents results for conditional double sorts on size and ETE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ETE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ETE and short stocks with low ETE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ETE Quintiles					ETE Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.95 [3.47]	0.90 [4.36]	0.90 [4.26]	0.82 [3.46]	0.71 [2.67]	-0.24 [-2.12]	-0.26 [-2.32]	-0.20 [-1.78]	-0.16 [-1.38]	-0.14 [-1.20]	-0.11 [-0.93]
	(2)	0.82 [3.67]	0.86 [4.37]	0.85 [4.11]	0.86 [3.77]	0.76 [3.07]	-0.06 [-0.72]	-0.14 [-1.74]	-0.07 [-0.95]	-0.05 [-0.59]	-0.07 [-0.94]	-0.05 [-0.68]
	(3)	0.72 [3.59]	0.74 [4.10]	0.83 [4.19]	0.82 [3.91]	0.78 [3.34]	0.06 [0.62]	-0.04 [-0.39]	0.05 [0.53]	0.05 [0.54]	0.11 [1.28]	0.10 [1.18]
	(4)	0.65 [3.33]	0.71 [4.18]	0.78 [4.36]	0.72 [3.66]	0.79 [3.64]	0.14 [1.51]	0.07 [0.78]	0.15 [1.78]	0.12 [1.40]	0.21 [2.53]	0.18 [2.10]
	(5)	0.46 [2.88]	0.59 [3.83]	0.56 [3.51]	0.57 [3.26]	0.67 [3.51]	0.21 [1.89]	0.10 [0.90]	0.24 [2.47]	0.24 [2.43]	0.37 [3.96]	0.36 [3.73]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ETE Quintiles					ETE Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	316	320	322	320	318	23	28	31	30	26	
	(2)	102	102	102	102	102	51	53	53	53	52	
	(3)	77	77	77	77	77	94	96	97	97	95	
	(4)	66	66	66	66	66	207	213	209	210	205	
(5)	62	62	62	62	62	1352	2025	1600	1711	1467		

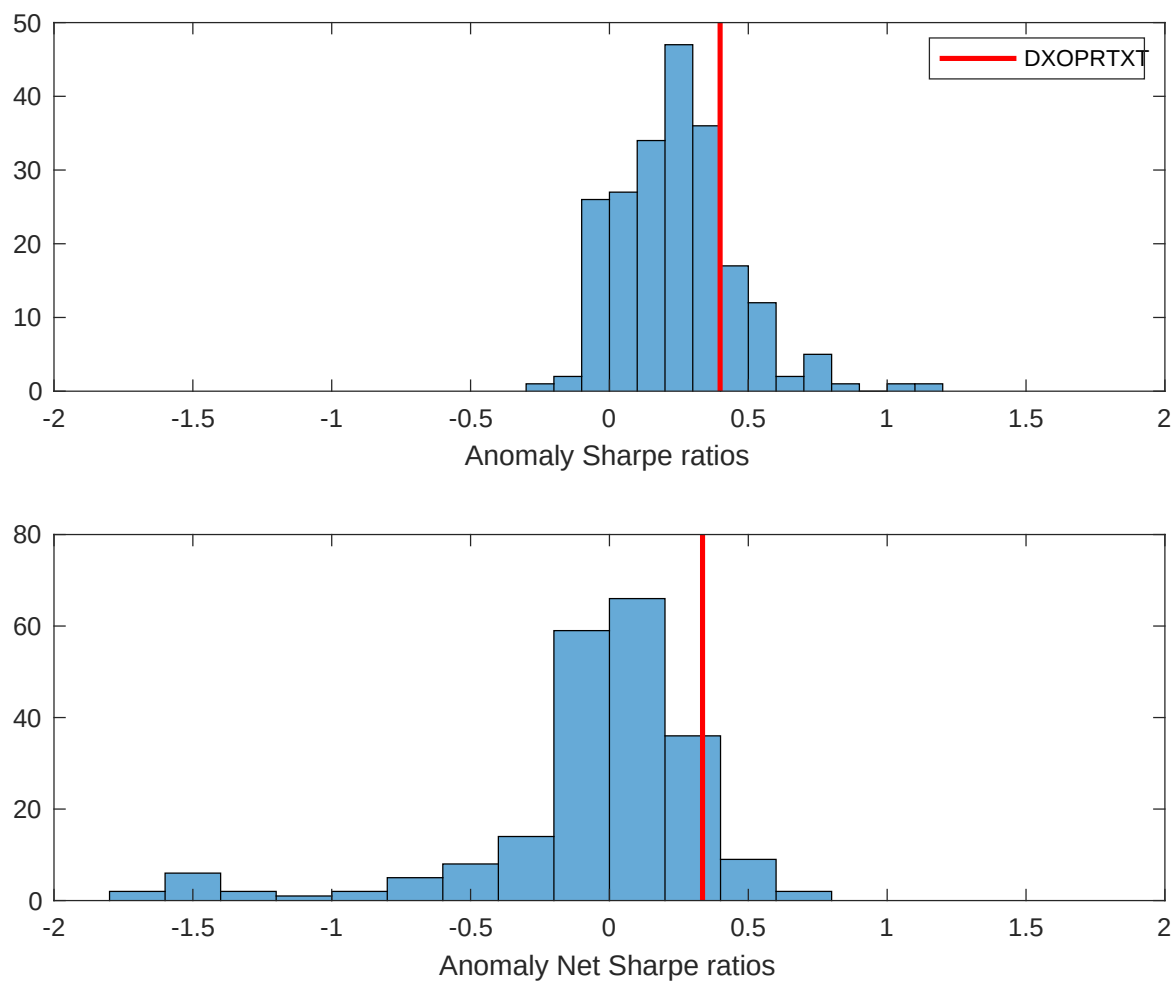


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ETE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

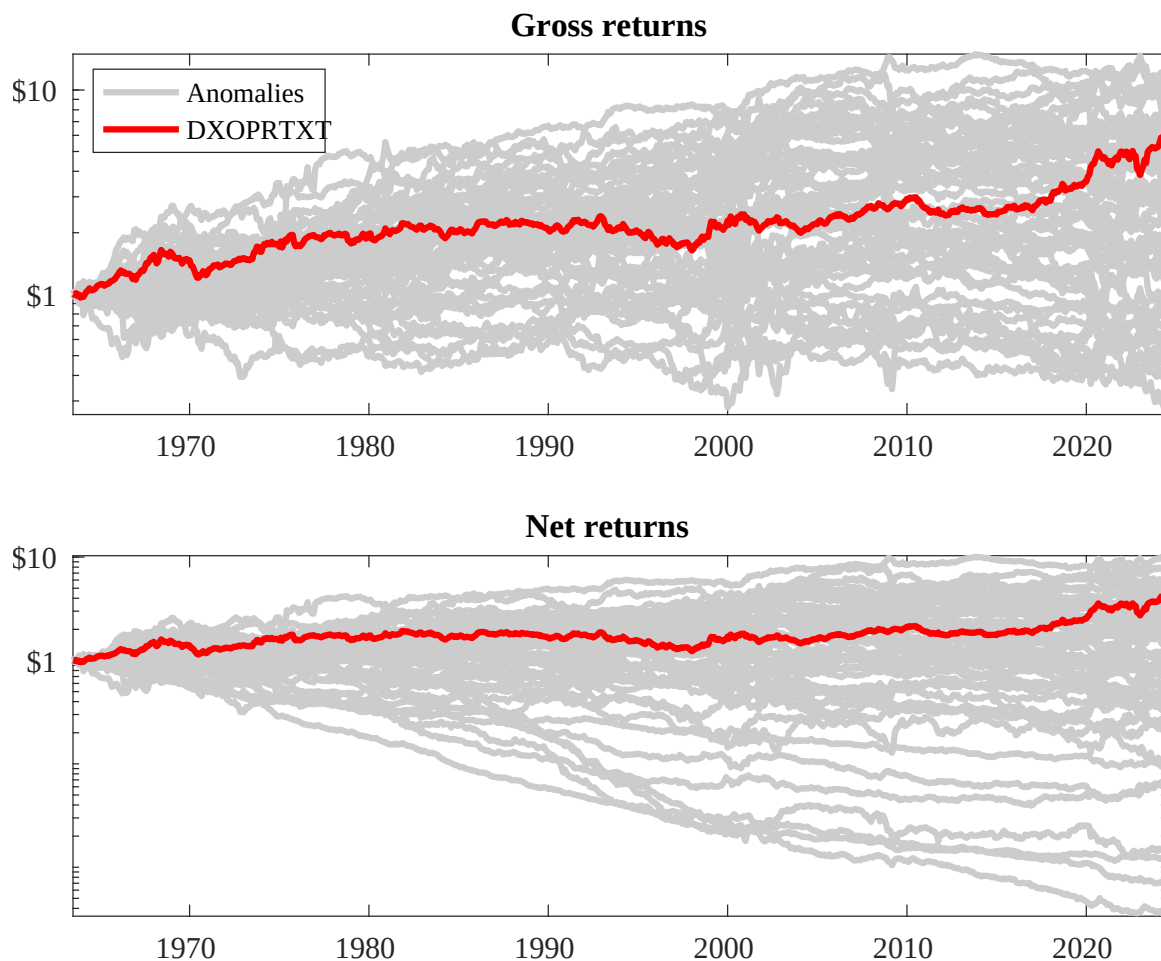
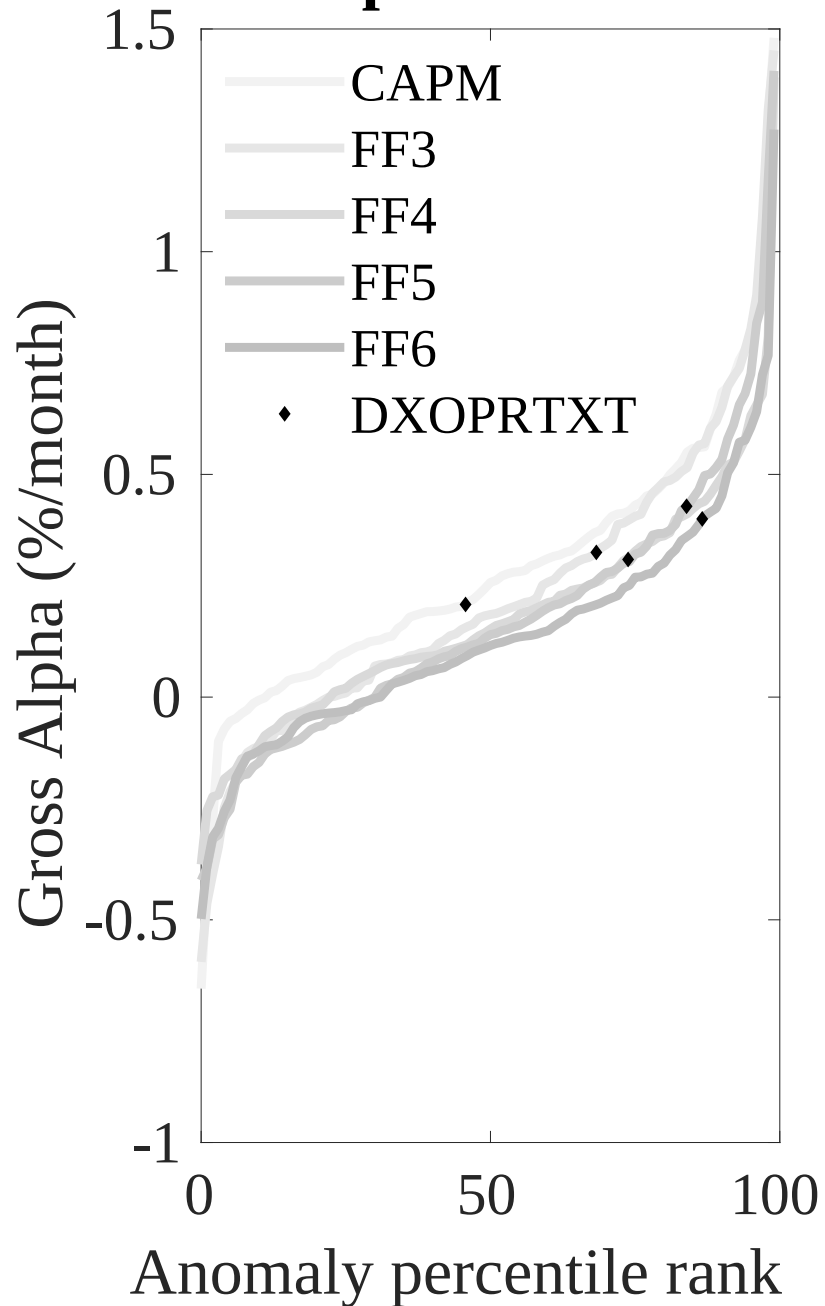


Figure 3: Dollar invested.

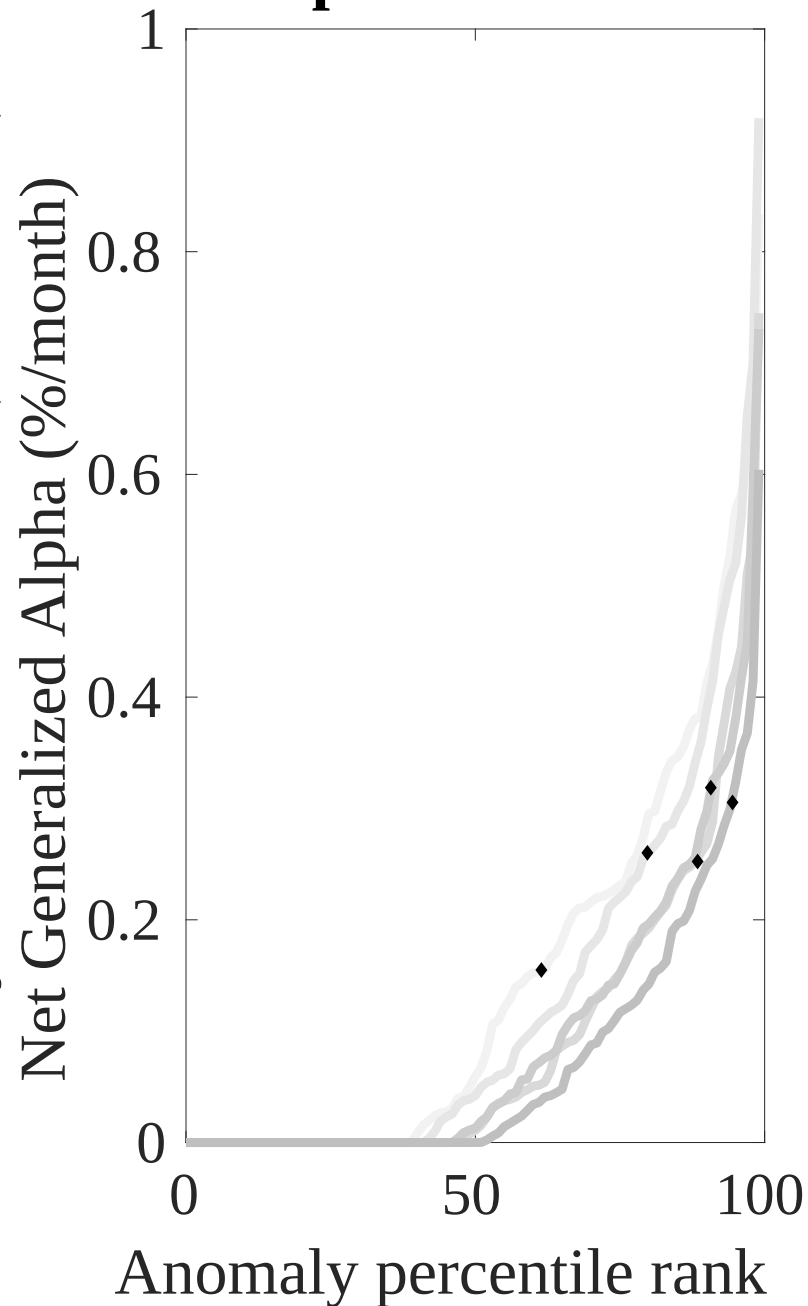
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ETE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



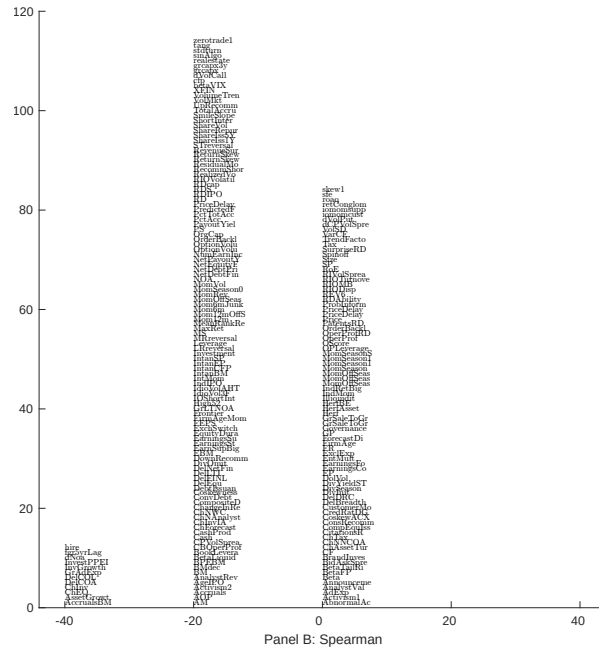
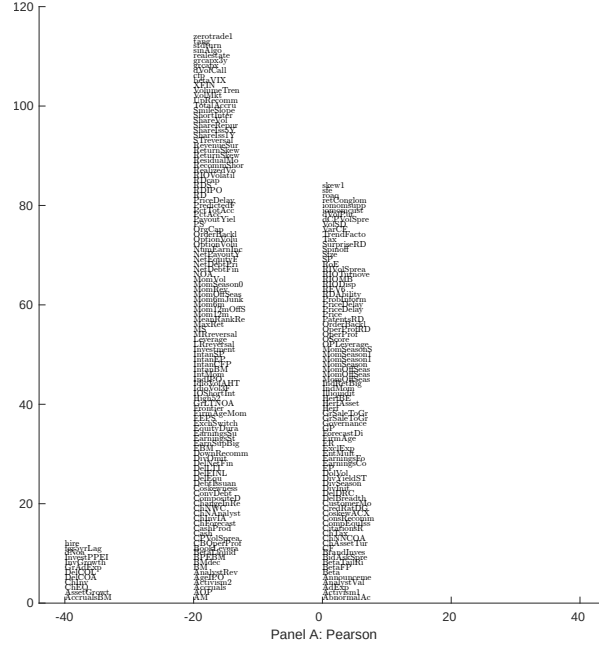


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ETE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

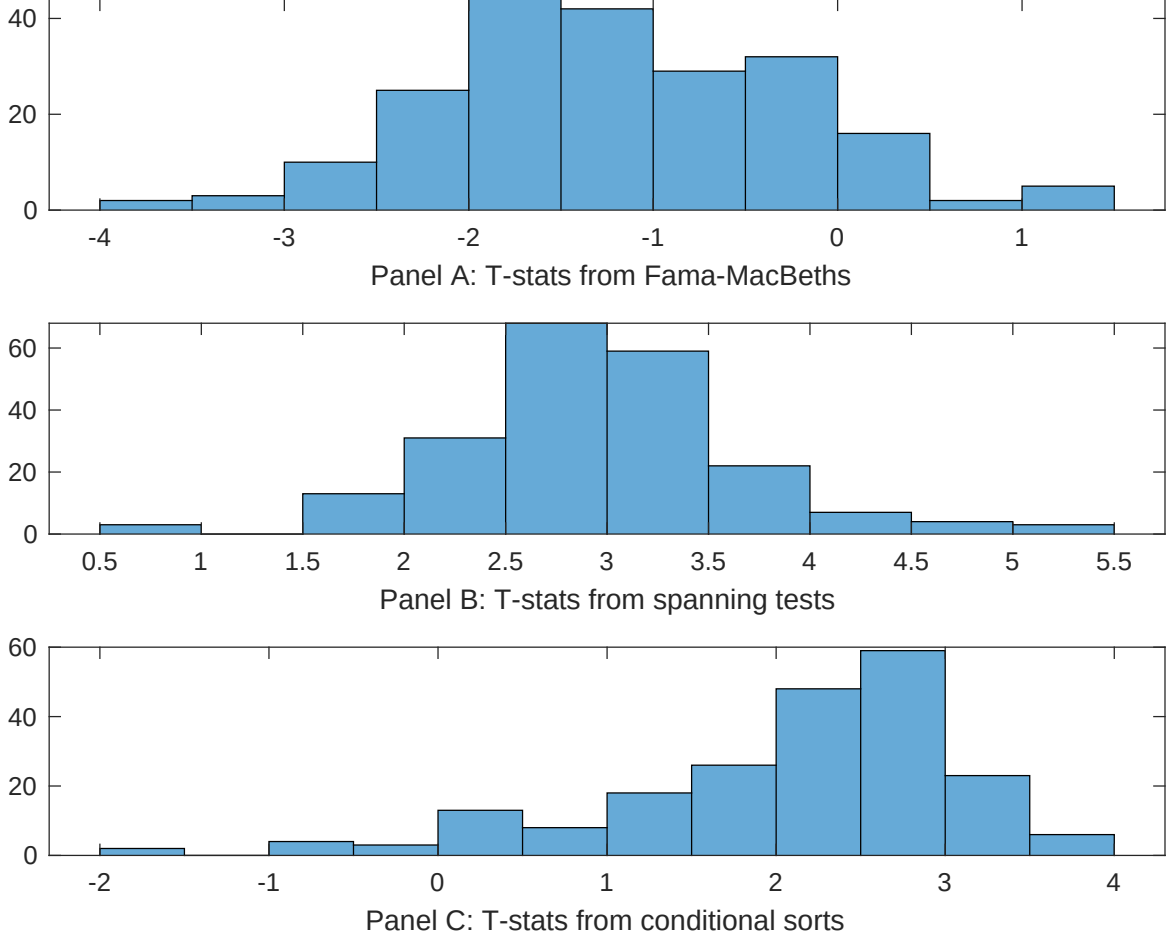


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ETE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ETE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ETE} ETE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ETE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ETE. Stocks are finally grouped into five ETE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ETE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ETE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ETE} ETE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.15]	0.12 [5.87]	0.13 [6.56]	0.13 [6.40]	0.14 [6.43]	0.12 [6.04]	0.13 [6.32]
ETE	-0.34 [-0.57]	-0.72 [-1.11]	-0.37 [-0.06]	-0.57 [-1.53]	0.42 [0.06]	-0.67 [-1.04]	-0.18 [-0.48]
Anomaly 1	0.18 [5.84]						0.10 [2.44]
Anomaly 2		0.83 [6.01]					-0.47 [-0.03]
Anomaly 3			0.96 [8.60]				0.46 [4.63]
Anomaly 4				0.76 [0.92]			-0.22 [-0.03]
Anomaly 5					0.15 [8.20]		0.16 [0.50]
Anomaly 6						0.14 [3.61]	-0.22 [-3.87]
# months	732	713	732	504	732	732	504
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ETE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ETE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.38 [4.85]	0.26 [3.30]	0.39 [4.85]	0.35 [3.78]	0.39 [4.93]	0.31 [3.94]	0.24 [2.84]
Anomaly 1	-25.80 [-6.67]						-8.53 [-1.77]
Anomaly 2		-42.09 [-10.05]					-34.65 [-6.44]
Anomaly 3			-13.49 [-2.80]				9.47 [1.87]
Anomaly 4				-12.99 [-4.13]			-2.41 [-0.80]
Anomaly 5					-25.12 [-6.80]		-19.72 [-4.88]
Anomaly 6						-24.96 [-6.67]	-7.20 [-1.45]
mkt	2.71 [1.44]	1.41 [0.77]	2.52 [1.31]	-3.95 [-1.63]	2.85 [1.51]	3.44 [1.82]	-1.11 [-0.51]
smb	8.99 [3.16]	12.47 [4.72]	15.45 [5.48]	3.59 [1.05]	15.37 [5.64]	13.63 [5.00]	2.65 [0.81]
hml	0.91 [0.23]	-1.82 [-0.50]	-8.35 [-2.23]	-13.49 [-3.07]	-6.04 [-1.65]	0.83 [0.21]	-2.89 [-0.66]
rmw	-1.30 [-0.35]	0.99 [0.28]	1.91 [0.51]	11.00 [2.55]	1.86 [0.51]	3.70 [1.00]	6.27 [1.60]
cma	-44.71 [-7.76]	-21.91 [-3.44]	-42.58 [-5.32]	-45.74 [-6.94]	-39.59 [-6.54]	-44.88 [-7.80]	-11.74 [-1.40]
umd	2.87 [1.53]	5.96 [3.27]	3.53 [1.84]	4.63 [2.19]	4.00 [2.14]	2.34 [1.24]	4.67 [2.43]
# months	732	713	732	504	732	732	504
$\bar{R}^2(\%)$	39	44	36	44	40	39	55

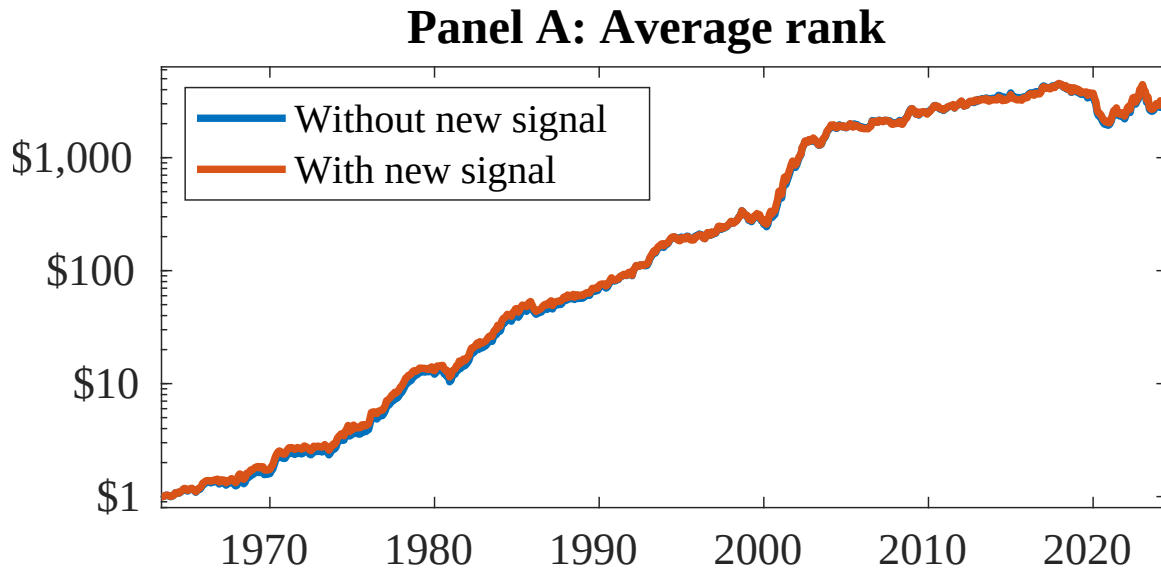


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ETE. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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